

# EMIR K ULUSOY

Staff RF / RAN Engineer · 5G NR, Open RAN, Capacity Planning · AI Systems in Telecom

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## PROFESSIONAL SUMMARY

- Eighteen years in wireless — 5G NR, 4G LTE, Wi-Fi and CBRS — across capacity planning for a national **300,000**-cell Open RAN network, network optimization, and device and RAN test. In the lab, in the field, and in the data behind both.
- Build and run AI systems in production: **multi-agent LLM orchestration** with LangGraph on AWS EKS, on-premise inference where data cannot leave the building, vision-language models applied to site photography, and the evaluation and observability that show whether any of it works.
- Led a team of **12 engineers** through 5G NR FCC and carrier approval programs. Protocol analysis from RRC and NAS through SIP and IMS to the RTP timing behind a voice call, with XCAL, XCAP, QXDM, Nemo, TEMS and Wireshark.
- **Patent** holder for LTE PIM detection (**US 10009784**) — a big data and machine learning method for finding passive intermodulation in network data instead of a site visit.
- Replaced licensed tooling with open-source equivalents in **Python** and **QGIS**: a **Linux** 5G/LTE/Wi-Fi throughput test tool (about **\$15K per license**), indoor and outdoor coverage reporting (about **\$5K per user**), a Planet TIFF-to-polygon converter (about **\$2K per user**), and field diagnostics parsers that cut roughly **\$100K a year** in repeat visits across **20+ field teams**.
- Looking for capacity, RAN engineering or **telecom AI** work where the domain knowledge and the systems side both matter, since that combination is what makes an AI system in telecom useful instead of a demo.

## PROFESSIONAL EXPERIENCE

### DISH Network, Littleton, CO

Staff Engineer — Wireless

Aug 2024 – Present

- Own capacity planning across network domains: traffic forecasting, headroom by layer, and the end-to-end model that flags a breakpoint before it is hit. Run 5G Open RAN capacity market by market, and set RAN capacity thresholds with the market teams that act on them, so upgrades land where customers feel the difference rather than where a fixed counter trips.
- Write the Python and SQL behind it: traffic and capacity forecasting across roughly **300,000 cells**, cell footprint and overshoot analysis across **40+** markets, and user-level capacity models validated against field measurements.
- Re-aggregated the **5G geolocation** pipeline onto a degree-based grid in AWS Redshift, matching stored precision to what GPS can actually resolve. Storage fell from **180 GB** a day to about **2 GB**, and the analysis window went from three weeks to two years — around **500M samples**, running on a laptop instead of a cluster.
- Review network KPIs at depth — accessibility, retainability, mobility, throughput — and trace degradations back to a cause: coverage, overshoot, interference or a parameter. Reconcile drive test against counters when the two disagree, since the field data is what settles it.
- Build the multi-agent, multi-channel orchestration for the company AI platform with **LangGraph on AWS EKS**, grounded in verified records with lineage kept, so any AI judgement traces back to its source. Own the observability and evaluation behind it: every request audited, silently swallowed errors captured, and scored question sets run against the deployed system rather than a local copy.
- Applied vision-language models on AWS Bedrock to before and after site photography, detecting equipment in the first and confirming removal in the second, replacing a manual step in closeout review.
- Present capacity, coverage and investment findings to **C-level stakeholders** quarterly — over **100 reports**, feeding decisions on programs worth **hundreds of millions**. Modeled incremental revenue against deployment cost to prioritize spend, returning 1.5:1 on the upgrades that went ahead.

### KCCTech, Walnut Creek, CA

R&D Director — RF Subject Matter Expert (from May 2021)

Dec 2017 – Apr 2024

- Led a cross-functional team of **12 engineers** through 5G NR testing and FCC and carrier approval programs for a national operator, covering test, analysis, fixes and reporting across **250+ site**, cluster and carrier approvals.

- Tested and validated 5G NR, LTE, Wi-Fi, Bluetooth, CBRS and GPS across **40+ RAN equipment types**, Apple and Android devices, modems, CPEs and AR/VR hardware, over the air in the field and in the RF box in the lab.
- Diagnosed RF and protocol stack anomalies on **30+ UEs** with QCAT, QXDM, Wireshark, XCAL, XCAP and TEMS: low throughput, handover and access failures, call drops, battery drain, overheating, MIMO and antenna performance. Validated fixes through lab retest and field trial.
- Debugged hardware, software and framework issues across L1 (PHY), L2 (MAC, RLC, PDCP) and L3 (RRC, NAS) alongside the teams that owned them. Analyzed VoNR, ViNR, VoLTE and CSFB performance from RRC and NAS through SIP and IMS to the RTP and RTCP timing behind the call.
- Advised 8+ **major clients** on 5G, LTE and Wi-Fi test methodology, and acted as the technical bridge between them and internal engineering when the two sides disagreed on a root cause. Ran UE certification with vendors, from the first failure report through to a compliant device.
- Scripted Open RAN lab testing with VNA for VSWR and S-parameters, VSA for EVM and IQ, oscilloscopes for jitter, and spectrum analyzers for interference. Located an external interferer at 668.5 and 682.5 MHz with a **sub-\$6K mapping setup**, against tens of thousands for a commercial equivalent.
- Owned client reporting in Power BI, Tableau, and SAP Business Objects, and the data validation behind it, so what a client saw in a report matched what was in the logs. Ran Wi-Fi 802.11a/b/g/n/ac/ax testing, including channel clarity validation across **20+ deployment scenarios**.

## Yupana Inc, Walnut Creek, CA

Senior RF Engineer

Aug 2014 – Nov 2017

- Built the LTE PIM catcher, a machine learning and big data method for detecting passive intermodulation from network data instead of a site visit. Granted as **US Patent 10009784**.
- Calculated **1,000+ LTE** link budgets for target cell-edge throughput under UL and DL noise rise conditions in Atoll.
- Automated the reporting chain around MS Office, SAP Business Objects and the drive test tools — TEMS Discovery, Genex Assistant, Nemo Analyzer — cutting data processing time by **70%**. On-site optimization for Digicel Haiti with Ericsson.

## Plano Engineering, Astana, KZ / Baku, AZ

RF Engineer

Jul 2009 – Jul 2014

- Ran RF engineering on Ericsson, Nokia and ZTE programs for Kcell, AzerCell and AVEA, including a nationwide frequency re-design and a Nokia to ZTE hardware swap.
- Conducted **75+ network benchmarks**, parameter trials and feature activations against contracted KPI targets.
- Performed radio network design in Atoll and resolved **300+** coverage, overshoot and interference problems in live networks.

## Egis Ltd, Ankara, Turkey

RF Engineer

Dec 2008 – Jul 2009

- RF device research and development, industrial automation design, and **FCC and CE** compliance research. Designed and built a **3G repeater**.

## EDUCATION

- Kocaeli University, Turkey — Bachelor of Science, Electronics Teaching Education (2003–2008). GPA 3.47/4.0. US academic equivalency: BS in Electronics Engineering (Park Evaluations, 2015). Highest ranking student in the department.

## TECHNICAL SKILLS

- **AI and ML:** LangGraph multi-agent orchestration, AWS Bedrock, AWS EKS, retrieval over verified records, vision-language models, local and air-gapped inference, model evaluation and observability.
- **Programming and data:** Python, SQL, JavaScript, Bash, Linux; Kiro, Amazon Quick; Snowflake, AWS Redshift, AWS Athena; pandas, GeoPandas; test case automation.
- **GIS and reporting:** QGIS, PyQGIS, MapInfo, iBwave, Atoll, Power BI, Tableau, SAP Business Objects.
- **RAN and protocol:** Open RAN, Ericsson ENM/OSS, Moshell, Samsung and Mavenir RAN, XCAL and XCAP (Accuver), QXDM, QCAT, Nemo, TEMS Investigation and Discovery, Actix, Wireshark, iPerf, OTA and RF box testing, VoLTE and VoNR analysis.
- **RF hardware and lab:** VNA, VSA, spectrum analyzer, oscilloscope, Site Master, antenna testing, PCB schematics, soldering, GPIO and I2C, Raspberry Pi, LabVIEW.